Solution of Problems of M670_8 (Hypothesis II)

Problem 1: A test on world history was given to a group of individuals before and also after a film on the history of the world was presented. The results are given below. We want to determine if the film significantly increased the test scores. Assume that the distribution of the paired differences (After - Before) is approximately normal.

Individual	After	Before
1	92	86
2	86	88
3	89	84
4	90	90
5	93	85
6	88	90
7	97	91

a. Give the hypotheses for this problem.

b. What type of hypothesis testing is this?

c. At 95% confidence (that is, 5% level of significance), test the hypotheses. (Which of the output below is correct?)

Solution

a. Hypotheses

We want to test if the film had a significant effect on the test scores. Specifically, we test whether the scores after watching the film are significantly higher than the scores before watching the film. The hypotheses are:

- **Null Hypothesis** (H_0):
$$H_0 : \mu_d = 0$$
where μ_d is the mean of the differences between the "after" and "before" scores.

- **Alternative Hypothesis** (H_1): The film significantly increased test scores. Mathematically:
$$H_1 : \mu_d > 0$$

b. Type of Hypothesis Testing

This is a **paired sample t-test** (or dependent t-test) because the test scores are measured from the same individuals before and after the film.

c. Testing the Hypotheses

1. **Calculate the differences** between "after" and "before" scores:

$$d_i = \text{After}_i - \text{Before}_i$$

For each individual:

$$d_1 = 92 - 86 = 6$$
$$d_2 = 86 - 88 = -2$$
$$d_3 = 89 - 84 = 5$$
$$d_4 = 90 - 90 = 0$$
$$d_5 = 93 - 85 = 8$$
$$d_6 = 88 - 90 = -2$$
$$d_7 = 97 - 91 = 6$$

The differences are: $6, -2, 5, 0, 8, -2, 6$.

2. **Calculate the mean and standard deviation of the differences**:

- Mean of differences ($\bar{d}$):

$$\bar{d} = \frac{6 + (-2) + 5 + 0 + 8 + (-2) + 6}{7} = \frac{21}{7} = 3$$

- Standard deviation of differences (s_d):

$$s_d = \sqrt{\frac{\sum(d_i - \bar{d})^2}{n-1}}$$

where $n = 7$.

Calculate $(d_i - \bar{d})^2$:

$$(6-3)^2 = 9$$
$$(-2-3)^2 = 25$$
$$(5-3)^2 = 4$$
$$(0-3)^2 = 9$$
$$(8-3)^2 = 25$$
$$(-2-3)^2 = 25$$
$$(6-3)^2 = 9$$

Sum of squares:

$$9 + 25 + 4 + 9 + 25 + 25 + 9 = 107$$

Standard deviation:

$$s_d = \sqrt{\frac{107}{6}} \approx 4.23$$

3. **Calculate the t-statistic**:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{3}{4.23/\sqrt{7}} \approx \frac{3}{1.60} \approx 1.88$$

4. **Determine the critical t-value**: For a one-tailed test at $\alpha = 0.05$ with $df = 6$, the critical value from the t-distribution table is approximately 1.943.

5. **Compare the calculated t-value with the critical t-value**: - Calculated t value: 1.88 - Critical t value: 1.943

   Since $1.88 < 1.943$, we fail to reject the null hypothesis at the 5% level of significance.

Conclusion

At the 95% confidence level, the test does not provide sufficient evidence to conclude that the film significantly increased the test scores.

Using Minitab, See Figure 1:

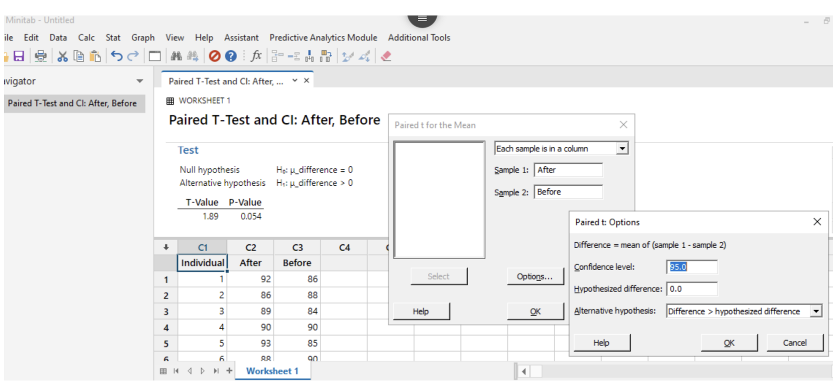

Figure 1: Minitab instruction for paired sample t-test.

Problem 2: Zip, Inc. manufactures Zip drives using two different processes. We want to determine if Process 1 takes significantly less time to manufacture the Zip drives compared to Process 2. Assume that the manufacturing times for both processes are approximately normally distributed. The data are as follows:

	Process 1	Process 2
Sample Size	27	22
Sample Mean (minutes)	10	14
Sample Variance (minutes^2)	16	25

Hypotheses

- **Null Hypothesis** (H_0): $\mu_1 \geq \mu_2$
- **Alternative Hypothesis** (H_1): $\mu_1 < \mu_2$

Test Statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Substituting the given values:

$$t = \frac{10 - 14}{\sqrt{\frac{16}{27} + \frac{25}{22}}} = \frac{-4}{\sqrt{0.5926 + 1.1364}} = \frac{-4}{1.315} \approx -3.04$$

Degrees of Freedom

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

$$df \approx \frac{2.99}{0.0135 + 0.0615} = \frac{2.99}{0.075} \approx 39.87 \approx 40$$

Critical Value For $df \approx 40$ and a one-tailed test at 95% confidence, the critical value of t is approximately -1.684.

Conclusion Since the calculated t-value (-3.04) is less than the critical value (-1.684), we reject the null hypothesis.

There is sufficient evidence at the 95% confidence level to indicate that Process 1 takes a significantly shorter time to manufacture the Zip drives compared to Process 2.

Problem 3: During the recent elections survey, the Republican presidential candidate showed the following pre-election voter support in Indiana and Alabama.

State	Voters Surveyed	Voters Favoring the Republican Candidate
Indiana	800	440
Alabama	600	360

1. We want to determine whether or not the proportions of voters favoring the Republican candidate were the same in both states. Provide the hypotheses.

2. Compute the test statistic.

3. Determine the p-value; and at 95% confidence, test this hypotheses.

a. Hypotheses

- **Null Hypothesis** (H_0): $p_1 = p_2$
  The proportion of voters favoring the Republican candidate is the same in both states.

- **Alternative Hypothesis** (H_1): $p_1 \neq p_2$
  The proportion of voters favoring the Republican candidate is different in the two states.

b. Compute the Test Statistic

- **Calculate the sample proportions:**

$$\hat{p}_1 = \frac{440}{800} = 0.55$$
$$\hat{p}_2 = \frac{360}{600} = 0.60$$

- **Calculate the pooled proportion:**

$$\hat{p} = \frac{440 + 360}{800 + 600} = \frac{800}{1400} = 0.5714$$

- **Calculate the standard error of the difference in proportions:**

$$\begin{aligned}
\text{SE} &= \sqrt{\hat{p}\,(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)} \\
&= \sqrt{0.5714 \times (1-0.5714)\left(\frac{1}{800}+\frac{1}{600}\right)} \\
&= \sqrt{0.5714 \times 0.4286\,(0.00125+0.00167)} \\
&= \sqrt{0.2459 \times 0.00292} \approx \sqrt{0.000718} \approx 0.0268
\end{aligned}$$

- **Calculate the z-statistic:**

$$\begin{aligned}
z &= \frac{\hat{p}_1-\hat{p}_2}{\text{SE}} \\
&= \frac{0.55-0.60}{0.0268} \approx \frac{-0.05}{0.0268} \approx -1.87
\end{aligned}$$

c. Determine the p-value and Test the Hypotheses

- **Find the p-value:**

$$\begin{aligned}
\text{p-value} &\approx 2 \times P(Z > 1.87) \\
&\approx 2 \times 0.0307 = 0.0614
\end{aligned}$$

- **Decision at 95% Confidence Level:** Since the p-value ≈ 0.0614 is greater than the significance level $\alpha = 0.05$, we fail to reject the null hypothesis.

Summary

- **Hypotheses**: $H_0 : p_1 = p_2$, $H_1 : p_1 \neq p_2$
- **Test Statistic**: $z \approx -1.87$
- **p-value**: ≈ 0.0614
- **Conclusion**: At the 95% confidence level, we do not have enough evidence to reject the null hypothesis. There is not a statistically significant difference in the proportions of voters favoring the Republican candidate between Indiana and Alabama.